

Deploying an Active Decoy (Honeypot) Lab

I recently completed a project focused on **Deception Technology** by deploying a Honeypot in a Linux environment to study attacker behavior and unauthorized access patterns.

The Concept

In cybersecurity, a **Honeypot** is a decoy system designed to look like a high-value target (like a web server or database). Its sole purpose is to be probed, attacked, or compromised so that defenders can gather intelligence on the "who, how, and when" of an intrusion.

The Build

Using **Pentbox** on **Kali Linux**, I configured a listener on **Port 80 (HTTP)**. This created a fake service that appeared open to anyone scanning my network.

Project Highlights:

- **Socket Listening:** Configured the system to "listen" for any incoming TCP handshakes on common attack ports.
- **Intrusion Logging:** The system was set to capture the **Source IP address**, **User-Agent string** (identifying the attacker's browser/OS), and the **Timestamp** of the attempt.
- **Deception Logic:** Provided a fake front-end to keep the "attacker" engaged while the system logged their activities in the background.

Results & Observations

During testing, the lab successfully:

1. Detected a unauthorized connection attempt from a remote device.
2. Logged a full trace of the "attacker's" fingerprint.
3. Triggered real-time visual alerts (the "Red Screen" alerts) for immediate SOC awareness.

Why This Matters

Honeypots are an essential part of **Threat Intelligence**. They help organizations identify "Bad Actor" IPs and understand the specific exploits being used in the wild, allowing for proactive firewall updates before the real production systems are targeted.

```

(kali@kali)-[~/Downloads/pentbox/pentbox-1.8]
$ sudo ./pentbox.rb
[sudo] password for kali:

PentBox 1.8

      .--.
     (oo)_____)--*
      (--)
      ||----||

----- Menu                      ruby3.3.8 @ x86_64-linux-gnu

1- Cryptography tools
2- Network tools
3- Web
4- Ip grabber
5- Geolocation ip
6- Mass attack
7- License and contact
8- Exit

  → 2

1- Net DoS Tester
2- TCP port scanner
3- Honeypot
4- Fuzzer
5- DNS and host gathering
6- MAC address geolocation (samy.pl)

0- Back

  → 3

// Honeypot //

```

```

// Honeypot //

You must run PentBox with root privileges.

Select option.

1- Fast Auto Configuration
2- Manual Configuration [Advanced Users, more options]

  → 1

HONEYPOT ACTIVATED ON PORT 80 (2026-02-24 12:49:46 +0530)

-----
INTRUSION ATTEMPT DETECTED! from 10.0.2.15:53480 (2026-02-24 12:50:39 +0530)
GET / HTTP/1.1
Host: 10.0.2.15
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Priority: u=0, i

-----
INTRUSION ATTEMPT DETECTED! from 10.0.2.15:54744 (2026-02-24 12:50:42 +0530)
GET /favicon.ico HTTP/1.1
Host: 10.0.2.15
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: image/avif,image/webp,image/png,image/svg+xml,image/*;q=0.8,*/*;q=0.5
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Referer: http://10.0.2.15/
Priority: u=6

```